

Obstacle Avoiding Robot Simulation using Arduino

Minor Project Report

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Abstract

Obstacle avoidance is one of the most fundamental capabilities required for autonomous mobile robots. This project presents the simulation of an obstacle avoider robot using Arduino Uno, an HC-SR04 ultrasonic sensor, an L293D motor driver IC, and two DC motors in the Tinkercad simulation environment.

The simulated robot continuously monitors the distance to obstacles using the ultrasonic sensor. When an object is detected within a predefined threshold distance of 20 cm, the Arduino processes the sensor data and executes an obstacle avoidance algorithm. The robot stops, reverses for a short duration, turns left, and then resumes forward movement.

The project demonstrates the integration of sensors, motor drivers, and embedded programming while providing an introduction to autonomous navigation concepts without requiring physical hardware.

Introduction

Autonomous mobile robots are widely used in industries, warehouses, hospitals, agriculture, and smart homes. One of the primary functions of these robots is to detect and avoid obstacles to ensure safe navigation.

Obstacle avoidance is achieved by continuously sensing the surrounding environment and making movement decisions based on sensor inputs. This project simulates an autonomous obstacle avoidance robot using Tinkercad, allowing the design and testing of the system in a virtual environment before physical implementation.

The robot uses an HC-SR04 ultrasonic sensor to measure the distance to nearby objects. Based on the measured distance, the Arduino Uno controls two DC motors through an L293D motor driver to avoid collisions automatically.

Objectives

The primary objectives of this project are:

- To simulate an autonomous obstacle avoiding robot using Tinkercad.
- To interface an HC-SR04 ultrasonic sensor with Arduino UNO.
- To control two DC motors using an L293D motor driver.
- To implement an obstacle avoidance algorithm.
- To understand the principles of sensor-based robotic navigation
- To gain experience with robotic simulation and embedded programming.

Problem Statement

Mobile robots operate in dynamic environments must detect obstacles and modify their movement to avoid collisions.

The challenge addressed in this project is to develop a simulated robot capable of:

- Detecting nearby obstacles
- Making autonomous movement decisions.
- Avoiding collisions without human interventions.
- Demonstrating the behaviour entirely within a virtual simulation environment.

Components used:

Component	Quantity	Purpose
Arduino UNO	1	Main Controller
HC-SR04 Ultrasonic Sensor	1	Distance Measurement
L293D motor driver IC	1	Motor control
DC Motors	2	Robot movement
Breadboard	1	Circuit Assembly
Jumper wires	As required	Connections
Tinkercad	N/A	Simulation Platform

Software used:

1. Tinkercad Circuits
2. Arduino IDE
3. GitHub (Documentation and Version control)

Working Principle:

The HC-SR04 ultrasonic sensor continuously emits ultrasonic pulses. These pulses travel through the air and reflect back after striking an object. The sensor calculates the distance using the time taken for the echo to return.

The Arduino continuously reads this distance and compares it with a threshold value.

If the distance is greater than 20 cm:

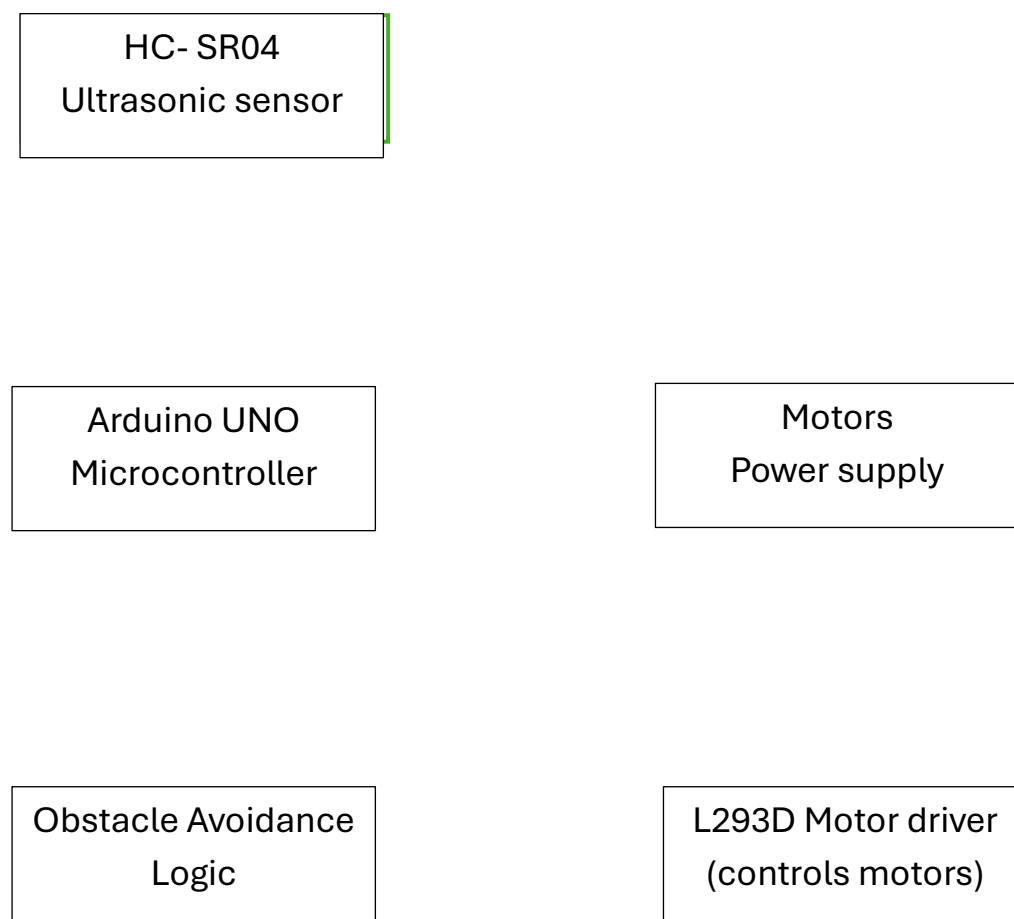
- The robot moves forward

If the distance is less than or equal to 20 cm:

- The robot stops.
- Moves back briefly.
- Turns left
- Continues moving forward

This cycle repeats continuously during the simulation.

System Architecture:



Circuit connections:

Ultrasonic Sensor:

Sensor Pin	Arduino Pin
VCC	5V
GND	GND
TRIG	D9
ECHO	D10

L293D Motor driver:

Arduino Pin	L293D Pin
D5	EN1
D2	IN1
D3	IN2
D6	EN2
D4	IN3
D7	IN4

Motors:

Left Motor- Output 1 & 2

Right Motor- Output 3 & 4

Algorithm:

1. Initialize Arduino
2. Configure sensor and motor pins
3. Read the distance from the ultrasonic sensor
4. Compare the measured distance with the threshold
5. If distance is greater than 20 cm move forward.
6. Else: stop, reverse, turn left.
7. Repeat continuously.

Testing Procedure:

The robot was tested by changing the obstacle distance in the Tinkercad simulation. Different obstacle distances were selected to verify the robot's response. The simulation confirmed correct behavior under different conditions.

Results:

Distance	Robot Actions
50 cm	Forward
35 cm	Forward
20 cm	Stop
15 cm	Reverse and turn
10 cm	Reverse and turn

Advantages:

- Easy to simulate without hardware
- Low- cost design
- Real- time obstacle detection
- Modular Architecture
- Simple programming logic
- Suitable for beginners in robotics
- Easy to modify and extend

Limitations:

- Simulation does not include real- world sensor noise
- Robot always turns left instead of selecting the best path
- Only front obstacle detection is implemented
- Does not detect object shape or type
- Limited navigation intelligence

Applications:

Obstacle avoidance robots are commonly used in:

- Warehouse automation
- Autonomous Guided Vehicles (AGV's)
- Smart vacuum-cleaning robots
- Delivery robots
- Educational robotics
- Industrial automation
- Research laboratories

Future Scopes:

Possible improvements include:

- Servo-mounted ultrasonic sensor for left-right scanning
- Dynamic path planning
- Multiple ultrasonic sensors
- Bluetooth control
- Wi-Fi monitoring using ESP32
- Camera based object detection
- Machine learning for advance navigation

Learning Outcomes:

This project helps in understanding:

- Arduino Programming
- Embedded systems fundamentals
- Ultrasonic sensor interfacing
- Motor driver operation
- PWM motor control
- Autonomous robot navigation
- Robotics simulation using Tinkercad
- GitHub based project documentation

Conclusion:

This project successfully demonstrates the simulation of an autonomous obstacle avoidance robot using Arduino Uno in the Tinkercad environment. The integration of the HC-SR04 ultrasonic sensor, L293D motor driver, and DC motors enable the robot to detect obstacles and execute simple avoidance manoeuvres automatically.

The project highlights fundamental concepts of robotics, embedded programming, sensor interfacing, and motor control while providing a safe and cost-effective platform for experimentation through simulation. It serves as a strong foundation for more advanced autonomous robotic systems involving intelligent navigation, multiple sensors, and wireless communication.

References:

1. Arduino Official Documentation: <https://docs.arduino.cc/>
2. Tinkercad circuits documentation: <https://www.tinkercad.com/>
3. HC-SR04 Ultrasonic sensor datasheet
4. L293D Motor Driver Datasheet
5. Arduino Language reference: <https://www.arduino.cc/reference/en/>